WHAT DOES WAVEFORM AUDIO ADD ONCE A SPEECH-TIME REFERENCE IS GIVEN? A FACTORIAL STUDY OF CROSS-MODAL TEMPORAL BINDING

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ABSTRACT

Speech indicates when to look but may describe a different moment. We cross speech-time reference availability with waveform access in a same-item factorial on 291 Cross-Modal Temporal Binding questions across six frozen audio-visual model configurations. With waveform retained, numeric references raise accuracy point estimates by 6.2–26.8 percentage points across four models, whereas localized references raise them by 26.5–34.4 points across two models. With the reference fixed, waveform audio lowers Qwen2.5-Omni accuracy by 15.1 points; the smaller ChronusOmni loss is correction-sensitive, whereas Qwen3-Omni remains inconclusive. Visual controls link recovery to referenced frames, while option-role, reversed-speech, and transcript probes characterize narration-associated answer shifts. On selected-speech CrossTask and COIN inputs, all exact paired tests are inconclusive and none of the joint interaction criteria is met. The factorial separately estimates reference gains and waveform effects under a fixed reference, without establishing a general audio-removal advantage.

***Index Terms*—** Audio-visual language models, speech timing, temporal binding, modality reliance

1. INTRODUCTION

Narration can preview or recap an action rather than describe the concurrent scene [**?**, **?**]. SVHalluc's Cross-Modal Temporal Binding (CMTB) task asks what is visible when speech occurs [**?**]. The choices target current action y, off-time narrated action d_n, and unrelated action d_u; some distractor roles are ambiguous. The waveform both marks when speech occurs and supplies content that may suggest another answer (Fig. **??**).

Removing waveform audio from the answering input removes both its occurrence cue and its linguistic and acoustic content. SVHalluc studies modality removal, transcript augmentation, and speech localization separately; it does not cross answering-time waveform access with an externally supplied speech-time reference on the same items. We therefore cross a supplied speech-time reference with waveform access on identical questions. The reference-fixed comparison asks what waveform adds with that reference fixed; the audio-retained comparison asks what the reference adds while waveform remains available. Models, questions, and decoding remain unchanged.

For Qwen3, adding the reference while retaining waveform audio raises accuracy from 59.8% to 86.6%; removing waveform audio then raises the point estimate to 90.0%. Qwen2.5 behaves differently: with the reference fixed, adding waveform audio lowers accuracy by 15.1 percentage points. Because CMTB is a targeted stress test by construction, our contribution is not to show that off-time narration can mislead, but to isolate its incremental waveform effect from reference gain. Visual, waveform, and error-direction controls accompany this diagnosis, while selected-speech CrossTask and COIN tests delimit its external scope.

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2. RELATED WORK

Speech detection identifies speech activity [**?**], audio–visual synchronization examines temporal correspondence between the two streams [**?**, **?**], and narration alignment relates spoken descriptions to depicted events [**?**, **?**, **?**]. Hallucination evaluations probe modality reliance and evidence-use errors [**?**, **?**, **?**, **?**, **?**]. Missing-modality AVQA instead studies robustness when audio is unavailable [**?**]. Temporal grounding extracts or learns time information [**?**, **?**, **?**]. OmniHalluc-L [**?**] studies valid evidence bound to the wrong moment or modality in long-form video, whereas CMTB fixes the queried moment by speech occurrence. Read-then-regenerate extracts a model's temporal signal before re-querying [**?**]; contrastive decoding alters evidence use [**?**, **?**]. Our crossed inputs measure waveform's additional value after a speech-time reference is supplied. Interaction interpretation also depends on measurement scale [**?**]; we foreground actual accuracy changes.

3. EXPERIMENTAL DESIGN

3.1. Task, models, and inputs

Before evaluation, we divided the 339 official CMTB items into video-disjoint development and evaluation splits containing 48 and 291 items, respectively. We selected the prompts and VAD settings using only Dev-48. Each evaluation item comes from a distinct source video. All models are frozen and answer the unchanged three-choice questions using greedy decoding, with a maximum of 32 generated tokens except for Phi-4, which uses 16.

Silero voice activity detection (silero-vad v6.2.1) [**?**] uses 16-kHz audio, threshold .5, minimum speech and silence durations of 250 and 100 ms, and 30-ms padding. The interval from the first onset to the last offset defines the speech envelope $\hat{I}_s$, including internal gaps; extraction uses no questions, options, or labels.

Qwen2.5-Omni-7B, Qwen3-Omni-30B-A3B-Instruct [**?**, **?**], Qwen-backed ChronusOmni [**?**], and Phi-4-multimodal [**?**] receive full video and a *numeric reference*: "The relevant speech activity occurs from START to END seconds. Answer using only the visual action during that interval. Do not infer or use transcript content." This numeric reference combines timestamps with visual-only instructions; both are held fixed in the reference-present waveform comparison. Video-SALMONN-2 [**?**] and VideoLLaMA2.1-7B-AV [**?**], both Qwen2-backed, instead receive a *localized reference*,

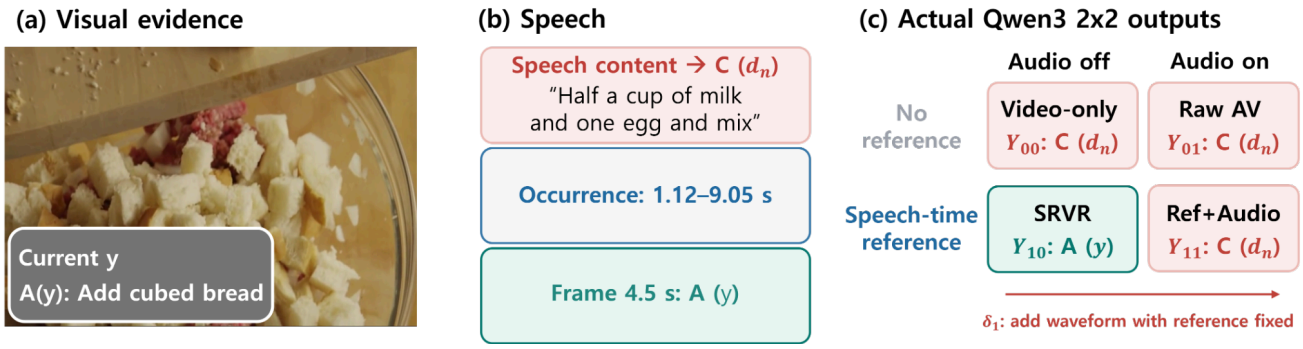


Fig. 1. Illustrative Qwen3 predictions across the four cells are C, C, A, and C, respectively. The δ_1 comparison adds waveform audio while holding the speech-time reference fixed. This item is illustrative; Qwen3's aggregate main-prompt effect remains inconclusive.

Table 1. CMTB-291 accuracies (%), reference effect with audio τ_1, reference-fixed waveform effect δ_1, and ceiling bound H (pp). Loss:gain and unadjusted exact p concern δ_1. The first four rows use numeric references; the final two use localized media. Effects and bounds are computed from unrounded proportions. †The full-cohort VideoLLaMA2.1 estimate is failure-sensitive: on 231 all-four-parsed items, $\delta_1 = +3.0$ pp (unadjusted $p = .189$).

Model	Y_{00}	Y_{01}	Y_{10}	Y_{11}	τ_1	δ_1	Loss:gain	Unadj. p	H
Qwen2.5-Omni	46.7	52.9	74.2	59.1	+6.2	-15.1	55:11	3.60×10^{-8}	+19.6
Qwen3-Omni	47.1	59.8	90.0	86.6	+26.8	-3.4	21:11	.1102	-2.7
ChronusOmni	47.1	68.4	86.6	80.8	+12.4	-5.8	28:11	.0095	-7.9
Phi-4	45.4	48.1	57.4	61.5	+13.4	+4.1	13:25	.0730	+39.9
Video-SALMONN-2	47.4	45.7	81.4	80.1	+34.4	-1.4	7:3	.3438	+20.3
VideoLLaMA2.1†	51.2	41.2	77.3	67.7	+26.5	-9.6	42:14	.0002	+32.6

$V[\hat{I}_s]$. Localized follow-ups crop video and audio together (Qwen: $\hat{I}_s \pm 1$ s; Chronus/Phi: exact $\hat{I}_s$), rebasing times. These are distinct intervention classes; we report model-specific effects and do not pool them. Audio-off uses native omission; Chronus uses zero tensors and omits its speech token.

Qwen samples at 2 fps; SALMONN-2 uses about 1 fps; Chronus, Phi, and VideoLLaMA2.1 use 64, 32, and eight frames. For Qwen3, after disabling 30-s truncation, we audited 1,164 item–condition records across four audio-on conditions (shared Raw AV plus the main, bare-timestamp, and localized reference conditions). We reran 386 records whose reconstructed processor tensors differed and conditionally reused 778 whose reconstructed tensors were bit-identical; historical tensors were unavailable.

3.2. Factorial and inference

Let $Y_{ra} \in [0, 1]$ denote accuracy, where r and a indicate reference and waveform presence, respectively. Tables report accuracies as percentages and differences in pp. The cells are Video-only (Y_{00}), Raw AV (Y_{01}), Speech-Referenced Visual Re-query (SRVR; reference present, audio off; Y_{10}), and Reference+Audio (Y_{11}). We report audio effects without/with reference, and the reference effect with audio:

$$\delta_0 = Y_{01} - Y_{00}, \quad \delta_1 = Y_{11} - Y_{10}, \quad \tau_1 = Y_{11} - Y_{01}. \tag{1}$$

The interaction is $\Delta_{\text{int}} = \delta_1 - \delta_0$. If waveform becomes redundant ($\delta_1 = 0$), the interaction is already $-\delta_0$. With the other three cells fixed, $Y_{11} \leq 1$ gives

$$\Delta_{\text{int}} \leq H = (1 - Y_{10}) - \delta_0. \tag{2}$$

Thus a negative interaction alone establishes neither harm nor an internal timing/content mechanism.

Full-cohort analyses score unparseable outputs and runtime failures as incorrect. Post-hoc pairwise complete-case sensitivities retain only items valid in both compared cells and do not replace full-cohort estimates. For the reference-fixed comparison, a loss is a correct-to-incorrect change after adding waveform audio; a gain is the reverse. Tables report loss:gain discordances and two-sided exact McNemar p-values [**?**]. CMTB/control CIs use 20,000 paired-source percentile bootstrap draws. For sparse external effects we additionally use conservative 95% bounds: subtract opposite endpoints of 97.5% Clopper–Pearson intervals for paired gain and loss probabilities [**?**]. For each model–dataset effect, Bonferroni gives at least 95% joint coverage over the gain and loss components. These wider bounds are reported as sensitivity intervals and do not replace the original decision rules.

After inspecting the 12 cells available for Qwen2.5, Qwen3, and Chronus, we post-specified a nine-test Holm family [**?**] containing δ_0, δ_1, and the logit interaction for each; inputs were fixed and no cell was rerun or retuned. Phi-4 and localized grids became available later as separate scope checks. Interactions use source-clustered generalized estimating equations (GEE; exchangeable correlation, robust errors) [**?**]. The original two-test bare family covers the logit interactions; a separate post-hoc two-test family covers the bare δ_1 effects. Other families cover four localized interactions, two planned content contrasts, and external tests; expanded checks and additional transcript analyses are post hoc. The two dataset-wise Qwen criteria each require both 97.5% Bonferroni interaction intervals below zero; the Chronus/Phi criterion requires all four 98.75% intervals below zero. Unadjusted p-values are descriptive.

3.3. Controlled external data

CrossTask [**?**] retains 93 of 117 sources (16 tasks; 15 non-cooking sources), disjoint from SVHalluc and a prior 132-source CrossTask pool. COIN [**?**] retains 85 of 190 candidates (23 tasks; 12 domains; 65 non-cooking sources), with zero overlap against 2,420 previously used source IDs. Two media-clock failures were excluded before inference without replacement. The CrossTask parent pool had been inspected for another task; final inclusion used availability and geometry, not target predictions.

Eligible VAD turns last 0.5–12 s, overlap exactly one action by $\geq 80\%$ of *turn duration*, and have a midpoint at least 1 s inside it. We select the longest eligible turn (ties: earliest start, then step ID) and the nearest two distinct non-target steps intersecting the context as distractors (interval distance to the speech midpoint; ties: step ID); option positions rotate cyclically. We center a window of length $\min(90\,\mathrm{s}, \text{video length})$ on the turn midpoint and clamp it to the video boundaries. We retain only the selected turn's audio at its original position and set all other audio samples to zero. Temporal overlap does not establish semantic agreement or conflict.

Qwen uses 2 fps (192-frame cap); Chronus/Phi use 64/32 timestamped frames. Within each model, visual tensors are matched across cells and the audio-on tensors across reference states. All 2,848 final external responses parse.

4. RESULTS

4.1. What does the reference recover?

For the three central numeric-reference models, adding the reference with waveform audio absent substantially raises accuracy. With waveform retained, gains range from 6.2–26.8 pp for numeric references and 26.5–34.4 pp for localized references (Table **??**). The four cells also permit arithmetic path decompositions, but these do not identify independent internal causal components.

With waveform audio absent, the following controls test whether recovery depends on visual evidence at the referenced time. Moving timestamps to the farthest equal-duration interval reduces Qwen3 accuracy from 90.0% to 4.5% and Qwen2.5 accuracy from 74.2% to 18.2% (Fig. **??**c). On 194 eligible matched-duration items, SALMONN-2 scores 80.9% at speech time versus 18.6% with non-overlapping random crops (mean of three seeds; difference CI $[55.5, 69.1]$ pp). At the same eight-frame budget, speech-time sampling raises vision-only InternVL2.5 [**?**] accuracy from 49.1% to 83.8%. Black video reduces Qwen2.5 and Qwen3 accuracy to 32.6% and 34.7%, respectively. Cyclic option rotation changes SALMONN-2 accuracy by +0.7 pp ($p = .85$).

4.2. What does waveform add with the reference fixed?

Under the post-specified central nine-test Holm family, reference-fixed waveform audio lowers Qwen2.5 and Chronus accuracy by 15.1 and 5.8 pp, respectively (adjusted $p = 2.16 \times 10^{-7}$ and .028); Qwen3's 3.4-pp estimate remains inconclusive (adjusted $p = .126$). Qwen2.5 survives the 12- and 18-test sensitivity families, whereas Chronus does not.

In the post-hoc two-test Holm family over the bare-timestamp δ_1 effects, both Qwen waveform losses remain significant (adjusted $p = 2.75 \times 10^{-9}$/.000346, respectively; Table **??**). Thus the full instruction wording is not necessary for the observed pattern, although Qwen3's main-prompt full-cohort result remains inconclusive. Without timestamps, the same audio-off visual instruction reaches only 45.0%/46.7% for Qwen2.5/Qwen3.

The central interactions are -21.3, -16.2, and -27.1 pp for Qwen2.5, Qwen3, and Chronus. Qwen2.5 retains a positive ceiling bound, whereas ceiling constraints force a negative interaction sign for Qwen3 and Chronus even at perfect Y_{11}. Significant logit interactions therefore do not by themselves distinguish reduced benefit from actual harm.

4.3. Which errors accompany waveform losses?

An annotator blind to model predictions labeled the fixed pre-correction 184-item Qwen error union; a second independently labeled a fixed random 100-item subset (78% agreement; $\kappa = .71$). Consensus plus single labels resolved 110 distractor-role assignments. Correct answers require no annotation, while ambiguous, unannotated, and failed responses remain residual. Narrated rates count confirmed d_n selections over all 291 items using this fixed, nonrepresentative map.

Qwen3 has 21 correct-to-incorrect transitions. Of these, 18 have determinate distractor roles, and 17 of those select d_n. Direct $y \to d_n$ and $d_n \to y$ transitions are 17:2 (exploratory exact $p = 7.29 \times 10^{-4}$). Using all 291 items as the denominator, the increase in confirmed narrated-choice rate exceeds the increase in confirmed unrelated-choice rate by 5.5 pp for Qwen3 and 12.4 pp for Qwen2.5. Even under adverse assignments of unmapped errors, these differences remain +5.2 and +10.3 pp, respectively. This supports error direction, not complete causation.

A Qwen2.5 reference-by-waveform probe reverses Demucs vocals segmentwise [**?**], matches root-mean-square amplitude, and restores background. A prediction-blind media/processor check retains 249 items. Without reference, reversed audio improves accuracy by 10.8 pp; with reference it lowers accuracy by 6.4 pp (CI $[-10.0, -3.2]$). Original audio costs a further 9.2 pp relative to reversed ($[-14.5, -4.4]$, post hoc). Original-only versus reversed-only narrated choices are 25:0 on 242 role-resolved pairs; correct responses need no distractor label, so this denominator can exceed 184. Because reversal changes both intelligibility and acoustic structure, the original-versus-reversed difference is not a pure semantic effect.

Appending original automatic-speech-recognition (ASR) text to audio-off SRVR lowers Qwen2.5 from 74.2% to 64.3% and Qwen3 from 90.0% to 80.8% ($N = 291$). On 244 fixed donor-eligible items, original versus unrelated text raises narrated choices by 14.3/8.2 pp (Qwen2.5/Qwen3; post-hoc two-test Holm $p < 2\times10^{-6}$). The planned, label-informed compatible-versus-unrelated donor comparison remains nonsignificant ($p = .27/.25$). Text can redirect answers, but its effect does not identify the waveform pathway.

4.4. External scope

On the selected-speech CrossTask and COIN grids, no unadjusted exact paired test is significant (all $p \geq .125$), and all conservative 95% bounds include zero (Table **??**). These results establish neither a consistent audio-removal advantage nor equivalence to zero.

Negative interaction is distinct from audio-removal benefit. CrossTask Qwen3 has $H = -9.7$ pp; COIN Qwen3/Chronus have $+9.4/+8.2$ pp. None of the three joint criteria is met. Turn selection, zeroing, and centered context limit interpretation; geometry does not establish semantic alignment. Chronus/Phi miss speech-interval frames in 3/15 CrossTask and 4/12 COIN items, though all target actions are sampled; excluding them leaves all four exact tests nonsignificant ($p \geq .25$), so full cohorts remain primary.

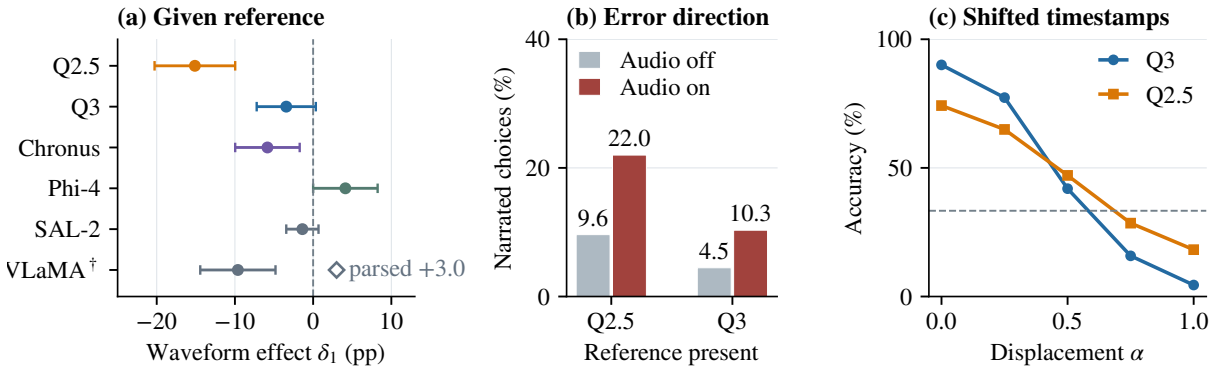


Fig. 2. (a) Reference-fixed waveform effects with unadjusted paired-bootstrap 95% CIs; the open diamond marks the VideoLLaMA2.1 all-four-parsed sensitivity ($N = 231$, $\delta_1 = +3.0$ pp). (b) Confirmed narrated-choice rates over all 291 items. (c) Audio-off accuracy as timestamps move from the detected speech interval ($\alpha = 0$) to the farthest equal-duration interval ($\alpha = 1$); the dashed line marks chance.

Table 2. CMTB-291 prompt/media controls. Bare uses only "Speech occurs from START to END seconds." CIs are unadjusted paired-source bootstrap intervals; simple-effect decisions use exact McNemar tests.

Reference	Model	Y_{10}	Y_{11}	δ_1 [95% CI]	Loss:gain	Unadj. p	Δ_{int}
Bare timestamp	Qwen2.5-Omni	78.0	61.2	-16.8 [-22.3, -11.7]	59:10	1.38×10^{-9}	-23.0
Bare timestamp	Qwen3-Omni	86.9	78.4	-8.6 [-13.1, -4.1]	36:11	.0003	-21.3
Localized	Qwen2.5-Omni	80.4	63.2	-17.2 [-23.0, -11.3]	68:18	5.29×10^{-8}	-23.4
Localized	Qwen3-Omni	78.7	73.5	-5.2 [-10.0, -0.3]	35:20	.0581	-17.9
Localized	ChronusOmni	80.1	71.8	-8.2 [-13.1, -3.4]	38:14	.0012	-29.6
Localized	Phi-4	66.0	61.5	-4.5 [-10.0, +1.0]	40:27	.1421	-7.2

Table 3. Selected-speech external δ_1 estimates (pp), conservative 95% bounds, loss:gain, and unadjusted exact McNemar p.

Model	δ_1 [95% interval]	$L{:}G$	p
CrossTask ($N = 93$)			
Q2.5	-7.5 $[-22.6, +8.3]$	14:7	.189
Q3	$+5.4$ $[-6.8, +16.9]$	3:8	.227
Chronus	$+1.1$ $[-11.9, +14.0]$	6:7	1.000
Phi-4	-4.3 $[-11.7, +3.6]$	4:0	.125
COIN ($N = 85$)			
Q2.5	$+1.2$ $[-13.8, +16.0]$	7:8	1.000
Q3	$+1.2$ $[-11.1, +13.3]$	4:5	1.000
Chronus	$+5.9$ $[-8.3, +19.4]$	4:9	.267
Phi-4	$+4.7$ $[-4.0, +12.7]$	0:4	.125

4.5. Robustness and sensitivity analysis

The speech envelope spans a median 11.96 s (45.0% of video). Silero thresholds .3/.5/.7 yield 72.2/74.2/72.5% Qwen2.5 audio-off SRVR accuracy ($-2.1/0/-1.7$ pp relative to .5); both outer paired CIs include zero. Whisper small.en [**?**], run via faster-whisper, gives 70.4% (-3.8 pp, 95% CI $[-6.9, -0.7]$).

Post-hoc pair-valid δ_1 estimates for Qwen2.5, Qwen3, and Chronus are -15.1, -4.5, and -5.9 pp at $N = 291$, 286, and 289. Their loss:gain counts are 55:11, 21:8, and 28:11 (unadjusted exact $p = 3.60 \times 10^{-8}$, .024, and .0095); these output-conditioned sensitivities do not replace the full-cohort estimates, and Qwen3 remains inconclusive. All-four-parsed interactions remain negative ($N = 283$, 283, and 289).

For VideoLLaMA2.1, 231 all-four-parsed items give $\delta_1 = +3.0$ pp ($p = .189$) and interaction $+6.1$ ($[0.4, 11.7]$). The sign reversal from the full cohort shows that this estimate is failure-sensitive rather than a stable waveform loss. On Temporal Alignment and Forecasting tasks ($N = 570$), replacing Raw AV with audio-free localized support changes SALMONN-2 accuracy by +1.2 pp ($p = .46$). Overall, the sensitivity analyses preserve the Qwen2.5 direction but leave smaller or failure-sensitive effects unresolved.

5. CONCLUSION

Reference interventions raise audio-on accuracy point estimates, but waveform effects vary: Qwen2.5 loses 15.1 pp, Qwen3's main-prompt full-cohort result is inconclusive, and external grids do not establish a general removal benefit. The repeatable finding is model-specific: Qwen2.5 accuracy declines across main, bare-timestamp, and localized references, whereas effects in the remaining configurations are heterogeneous. The crossed design separates reference availability from waveform access; ablation gains alone do not prove that audio is harmful. The contribution is diagnostic: it identifies the comparison needed to estimate waveform's added effect under a fixed reference while showing that negative interactions and audio-off gains are not interchangeable. Post-specified analyses, conditional reuse, partial labels, and selected-speech construction limit inference. Future work should pre-specify correction families, separate timestamps from modality-use instructions, and evaluate semantically verified narration conflicts with open-ended responses.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzed publicly released audiovisual benchmark data and annotations created by members of the research team. No human participants were recruited as research subjects, and ethical approval was not required under the applicable institutional policy.

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